

-
-
-
-
-

Chapter 4

Inventory Updates



Drinks Inventory Management Inventory Updates

In order to improve the speed and quality of information displayed within Drinks Inventory Management, a series of weekly and periodic balances are maintained. These balances need to be updated at week, month and year end.

At year end, week end, month end and year end, updates should be run consecutively. The month end run normally includes a valuation of stock movements and, therefore, is prompted.

All the updates, as well as global duty, standard cost transfer and expense account adjustments can be processed from the Drinks Inventory Updates menu.

Stock Management

The investment in Drinks Inventory typically represents one of the largest single areas of investment in a business, often as high as twenty-five percent of total assets. It is therefore vital that planning and control of its acquisition and disposition are exercised.

A number of techniques have been developed to assist in this control. In the sphere of a Production operation, where demand is considered to be 'dependent', the most common forms of control are: Material Requirements Planning, or Just In Time. These techniques work in the relatively predictable environment where demand is dependent on an agreed production schedule.

However, the demand for end-product items is primarily influenced by factors that are independent of company decisions. These external factors include a certain amount of random variation in the demand for such items. As a result, forecasts of demand for these items are typically projections of historical demand patterns to determine average usage. The concept of independent and dependent demand are important in selecting appropriate inventory management techniques.

There are generally considered to be two types of inventory: Movement and Organisational.

Inventory Holding

Movement Inventories

These exist purely because time is required to make and transport items. The most visible of these is work in progress in a Production organisation and stock in transit in a distribution environment.

Organisation Inventories

These exist to de-couple successive stages of production and distribution. Finished goods inventory makes it unnecessary to produce directly to meet the demands from distribution centres. This permits flexibility in scheduling Production and assembly operations. Organisational inventory can be held to meet different requirements; these should also be understood and are: cycle stock, safety stock and anticipation stock.

Cycle Stock

This is a result of buying or making more than is required for the immediate requirement. The one time costs of raising an order and perhaps setting machines or transport must be set against the costs of holding stock, to determine the 'Economic Order Quantity'.

Safety Stock

Is to protect against uncertainty. That is not only the uncertainty of demand, but also of the likelihood that planned lead times will be achieved.

Anticipation Stock

Is that which is held because there are times when demand exceeds the rate at which the item may be economically supplied. Highly seasonal products such as lawn mowers are a good example where it is appropriate to have a fairly steady rate of production throughout the year, and build high levels of stock over the winter so that the depletion rate of stock plus the Production rate equals demand during high summer.

Inventory Decisions

Two types of routine decisions need to be made in managing inventories. These decisions are concerned with how much should be ordered and when should the orders be placed. Four main factors influence these decisions:

- Forecast demand
- Replenishment lead time
- Inventory related cost
- Management policies

Two important parameters assist inventory controllers in making these decisions, they are:

Economic Order Quantity - EOQ

Economic Time Between Orders - ETBO

The standard calculation of EOQ is:

$$EOQ = \frac{\sqrt{2CP \times AU}}{CH}$$

Where:

CP = Cost of order placement

AU = Average usage per annum

CH = Inventory holding cost per year

$$CH = \frac{ItemCost * CarryingRate}{100}$$



The cost of order placement and carrying rate are both specified on the Inventory company profile.

Once EOQ has been determined then:

$$ETBO = \frac{EOQ}{WU}$$

Where:

WU = Average usage per week



The resulting Economic Time Between Orders (ETBO) is in weeks.

Drinks Inventory Management provides a range of facilities which contribute to the management of independent demand items which are traditionally controlled by reorder point techniques.

- A value/usage matrix enables a stock investment policy to be implemented at each stockroom
- Usage profiles provide a range of calculations for forecast usage

- EOQ calculation is performed which could be used as the basis of reorder quantity and time between orders
- Lead time can be factored in the reorder calculation to modify order review periods

Re-Order Policy Calculations

As discussed in the Maintenance section, there are three Inventory Re-order Policies:

- Reorder up to the maximum stock level
- Reorder the reorder quantity.
- Reorder up to the reorder point.

The reorder quantity (ROQ) is calculated in a different way for each policy:

1 - ROP/MAX

$ROQ = \text{Maximum Stock} - \text{Expected Stock} + (\text{Average Usage Lead Time})$

2 - ROP/ROQ

$ROQ = \text{Average Usage Lead Time Factor Percentage}$

3 - Up to ROP

$ROQ = \text{Reorder Point} - \text{Expected Stock} + (\text{Average Usage Lead Time})$

Where:

$\text{Maximum Stock} = \text{Max Stock Weeks Average Usage}.$

$\text{Reorder Point} = (\text{Min Stock Weeks} + \text{Lead Time (in weeks)}) \text{ Average Usage}.$

$\text{Expected Stock} = \text{Available} + \text{On Order} + \text{In Transit} - \text{Back Orders} - \text{Reserved}$

Economic Order Quantity is discussed earlier in this section.

Week End Update



1/DID

This batch function performs the following tasks:-

- If the usage cycle set at company level is 'W' for weekly, then average weekly usage will be recalculated for each item/stockroom.
- The physical and in transit stock at the end of the week is stored in a separate field 'Stock at start of week' for each item/stockroom.
- Usage this week is set to zero for each item/stockroom.
- The current week number on the calendar file is incremented by 1. If the end of year is reached then week number is set to 1.



Note: If processing the last week of the year, this update should immediately be followed by month and year end processing since the year number on the calendar file is only incremented by the year-end update.

Week End Processing

To process the week end update, simply take the option to submit the job. No selection parameters are required.

Month End Update

1... **2/DID**

This batch function performs the following tasks for each item/stockroom:

- If the usage cycle set at company level is 'M' for monthly, then the average monthly usage will be recalculated.
- Usage this period is set to zero.
- Number of sales orders this period is set to zero.
- Number of substitutions this period is set to zero.
- Current physical and in transit stock is stored in a separate field 'Stock at start of period.'
- The stock value is calculated using the appropriate costing method and stored in a field called 'Stock value at start of period.'

The function will also:

- Increment the current period number by 1. If the end of the year is reached the period number is set to 1.
- Produce a stock movements valuation report if this has been requested as part of the submission process. This report is produced based on parameters entered at that time.

Month End Update Selection Screen

- ➔ Select the Month End Update option and a screen is displayed which requires the parameters for the stock movements valuation report.

IN056		HW - Handy Wholesalers Ltd		User PGMR1		9/12/96	
						9:53:23	
Record Report Selection Criteria							
		From		To			
Stockroom code.....?	01			02			
Item.....?	B201			B999			
OR							
Item group major/minor.....?							
OR							
Item type.....?							
OR							
Item class.....?							
F3=Exit F4=Prompt 12-50 S MW KS IM II KENB20A PET007S2							

Enter the selection parameters or press **F3=Exit** if you have already produced this report for this period. A full explanation of this report can be found in the Reports section of this product information.

- ➔ Press **F8=Submit**, and a Month End Update Confirmation Screen, which includes the regeneration of ABC classifications, is displayed.

Month End Update Confirmation Screen



This screen is displayed when you have pressed **F8=Submit** from the Month End Update Selection Screen.

```
IN105                NR - Neil Radford and Co. Ltd      User GBJBA1MS      13/08/96
                                                           16:40:50
Month End Update Confirmation

Regenerate ABC classifications ..... 1

F3=Exit              F8=Submit

06-47      Sf      MW      KS      IM      II      KENB20A      PET166S2
```

Fields

Regenerate ABC Classifications

This option is given to regenerate ABC details.

1 - Generate.

0 - Do not generate.



Press **F8=Submit** to process the month end updates or **F3=Exit** to cancel.

Year End Update

1...
2...

3/DID

This batch function performs the following tasks:

- Allows the number of weeks this year and last year specified on the calendar to be updated.
- Allows the number of periods specified on the calendar to be changed.
- Sets any yearly accumulators to zero.
- Increments year number by 1.

Year End Update Calendar File Maintenance Screen



Select the Year End Update option.

The calendar file maintenance screen is displayed.

IN012	HW - Handy Wholesalers Ltd	USER PGMR1	9/12/96
			9:54:31
Maintain Calendar File			
Current year.....		86	
Current week.....		10	
Current period.....		2	
No. of weeks this year.....		52	
No. of weeks last year.....		52	
No. of periods this year...		12	
F3=Exit F11=Delete			
06-30	86	HW	KS IM II KENB20A PET087S2

Fields:

Current Year

Enter up to two numeric characters which specify the current year.

Current Week

Enter up to two numeric characters which specify the current week.

Current Period

Enter up to two numeric characters which specify the current period.

No Of Weeks This Year

Enter up to two numeric characters to specify the number of weeks this year. The value must be less than or equal to 53.

No Of Weeks Last Year

Enter up to two numeric characters which specify the number of weeks last year. The value must be less than or equal to 53.

No Of Periods This Year

Enter up to two numeric characters which specify the number of periods this year. The value must be less than or equal to 13.



Review and update the number of periods and

the number of weeks this year and last year so that they reflect the new year.



Press **Enter** to validate the details, then **F8=Submit** to send a batch job which will perform the year end update.

Calculation of Re-Order Point, Maximum Stock and EOQs

1... **4/DID**
2...

This procedure, calculates the Reorder Point, Maximum Stock, and EOQ for all stockroom balance records where it has been requested (See 'Stockroom Details - Maintenance').

For each item / stockroom, the procedure first determines the value/usage category by using the item cost, average usage and the stock weeks matrix. Then, using the minimum and maximum stockholdings specified for the category, the procedure calculates the quantities as follows:

Reorder Point =

(Minimum Stock Weeks + Lead Time Weeks) x Average Usage

Maximum Stock =

(Maximum Stock Weeks + Lead Time Weeks) x Average Usage

Economic Order Quantity =

$$\sqrt{\frac{2 * OrderPlacementCost * AnnualUsage}{InventoryHoldingCostPerYear}}$$

This procedure may be requested as an ad hoc update process, but would generally be run at period, or week-ends. No selection parameters are required, simply take the new option to submit a job and perform the required updates.



Note: Any stockroom balance records found without an associated item master or stockroom detail record will be written to an exception report.

Reorder / Overstock Report

This procedure produces a report which can be used to identify stock shortage or overstocking, according to the stock management information.

The stock position for all items is reviewed as detailed below:

- Available stock is calculated: $\text{Physical} - \text{Frozen} - \text{Allocated} = \text{Available}$
- Expected stock is calculated: $\text{Available} + \text{Order} + \text{In Transit} - \text{Back Order} - \text{Reserved} = \text{Expected}$
- If expected stock is greater than maximum stock, the item is reported as in Excess
- If expected stock is less than reorder point, the item is reported for Order Action
- If Order Action is required, the reorder quantity is determined. If a reorder quantity has been specified manually, it is recommended regardless of the reorder policy. If not, the reorder quantity is determined for each policy as follows:
 - ROP/MAX: the system recommends reorder up to the Maximum Stockholding
 - ROP/ROQ: the system calculates a reorder quantity based on the reorder quantity factor, the average usage and the lead time
 - Up to ROP: the system recommends reorder up to the reorder point.

Details of the Re-Order / Overstock Report can be found in the Reports section of the product information.

Calculate ROP, Max Stock Processing Window



Select the Calculate ROP, Max Stock option.

The standard batch processing window is displayed.

DID	HM - Handy Wholesalers Ltd	User PGMR1	09/12/96 09:55:08
Inventory Updates			
Regular Updating		Budget	
1. Week end update		21. Global duty increase	
2. Month end update			
3. Year end update			
Ad-Hoc Updating		Costing	
4. Calculate ROP, max stock etc.		24. New Standard Cost transfer	
Day End		Calculate ROP, max stock etc.	
5. Expense Account Ad		The option you have chosen submits a batch job to perform the requested task.	
		Press F8 to confirm this submission or press F3/F12 to return without submission.	
Option 4		F3=Exit F8=Confirm submit F12=Previous	
Search			
F3=Exit F4=Prompt F6=Messages F12=Previous F16=Change company F18=Prints			
21-11 52 MW KS IM II KENB20A PET087S2			



Press **F8=Confirm Submit** to send a batch job which will perform calculations, or press **F3=Exit** to exit the screen without submitting the job.

Expense Account Adjustments

1...
2...

5/DID

This batch job will produce a report on all the stockrooms that have been involved in an expense account transaction.

It will contain the same information as the Expense report in the Reports section. Refer to Expense Account Adjustment Report for more detail.

Expense Account Adjustment Processing Window



Select the Expense Account Adjustments option.

The standard batch processing window is displayed.

DID	HM - Handy Wholesalers Ltd	User PGMR1	09/12/96 09:55:50
Inventory Updates			
Regular Updating		Budget	
1. Week end update		21. Global duty increase	
2. Month end update			
3. Year end update			
Ad-Hoc Updating		Costing	
4. Calculate ROP, max stock etc.		24. New Standard Cost transfer	
Day end processing for Daily Stock Adjustments			
Day End		The option you have chosen submits a batch job to perform the requested task.	
5. Expense Account Ad		Press F8 to confirm this submission or press F3/F12 to return without submission.	
Option <u>5</u>		F3=Exit F8=Confirm submit F12=Previous	
Search _____			
F3=Exit F4=Prompt F6=Messages F12=Previous F16=Change company F18=Prints 21-11 SP MW KS IM II KENB20A PET087S2			



Note: When run as a day end process, the expense account movements that are reported are deleted from the database. A report is produced for each stockroom.



Press **F8=Confirm Submit** to send the report to print. Press **F3=Exit** to exit the screen without submitting the job to batch.

Global Duty Increases

1...
2...

21/DID

Whenever drink duty tariff changes, all the products that are affected need the change to be incorporated into the price list. The price lists are altered by examining the product type and updating item prices within a group by a specified rate. The changes will be incorporated in new orders if the price list is effective.

This is a vital option for the Advanced pricing system involved in the Drinks Distribution package. Without it, each price list of Drinks Sales Order Processing would need to be examined and updated for all the products that appear within the tariff code.

Global Duty Increases Amendment Screen



Select the Global Duty Increases option.

The screen displayed shows the existing tariff codes and the corresponding tariff rates.

IN091	HW - Handy Wholesalers Ltd	User PGMR1	9/12/96
			10:05:06
Global Duty Amendment			
Tariff code	New rate	Current rate	Description
- BI	.00	1.00	Bitter
- CD	.00	7.00	Carbonated Drinks
- CI	.00	1.00	Cider
- FJ	.00	1.00	Fruit Juices
- LG	.00	4.00	Lager
- MX	.00	1.00	Mixers
- SD	.00	1.00	Soft Drinks
- WI	.00	12.00	Wines-Red/White/Rose/Sparkling
- 01	.00	20.00	Brandy
- 02	.00	40.00	Whisky
- 03	.00	123.00	Vodka
- 04	.00	7.00	Rum
- 05	.00	75.00	Gin
- 06	.00	7.00	Vermouth
- 99	.00	21.00	Duty Free
-	.00	.00	
4=Delete from this run			
F1=Help F3=Exit F5=Reset F8=Submit job F12=Previous			
06-04	SA	MW	KS IM II KENB20A PET007S2

Fields:

Select

This field is untitled on the screen. Enter a '4' and press **Enter** to delete a tariff. You will need to press **Enter** again to confirm deletion

New Rate

To change the current rate, enter the new rate alongside the old, and press **Enter**.

Make changes to all the products that require a tariff alteration.



Tip: You can cancel any changes you have made before submitting the job, by pressing **F5=Reset**.



Note: This updates the duty rate on the Description file, updates the duty value on all items with the tariff code, and updates the item new standard cost and base price in duty paid stockrooms.

It also prepares a file for use by the global duty update function (see Drinks Sales Order Processing).



Press **Enter** to validate the new rates followed by **F8=Submit Job** to start the job of updating the

rates and the price lists within Drinks Sales Order Processing.

New Standard Cost Transfer



24/DID

The New Standard Cost including the effective Start Date of an item can be changed via Drinks Inventory Management Items Maintenance. If the date is current, you are asked to update, otherwise, this option can be used to perform the transfer. The update replaces Standard Cost with New Standard Cost for all depots that contain the item.



Note: For Bonded Stockrooms, the New Cost is the New Standard Cost minus the Item Duty Value.

New Standard Cost Transfer Window

 Select the New Standard Cost Transfer option.

The standard batch processing window is displayed.

DID	HW - Handy Wholesalers Ltd	User PGMR1	09/12/96 11:30:25
Inventory Updates			
Regular Updating		Budget	
1. Week end update		21. Global duty increase	
2. Month end update			
3. Year end update			
Ad-Hoc Updating		Costing	
4. Calculate ROP, max stock etc.		24. New Standard Cost transfer	
Day End			
5. Expense Account Ad			
'New Standard Cost transfer' The option you have chosen submits a batch job to perform the requested task. Press F8 to confirm this submission or press F3/F12 to return without submission. F3=Exit F8=Confirm submit F12=Previous			
Option	24		
Search			
F3=Exit F4=Prompt F6=Messages F12=Previous F16=Change company F18=Prints 21-11 S MW K IM II KENB20A PET087S2			

 Press **F8=Confirm Submit** to send the job to print. Press **F3=Exit** to exit without submitting the job.